Q1:
4 full cycles in 10 ns
1 full cycle in 10/4 = 2.5 ns = 2.5 x 10^{-9} s
This is the time period.

Frequency = 1/(time period) = 1/(2.5 x 10^{-9}) = 4 x 10^{8} Hz

According to the book, distortion occurs in composite signals. Although the phase is shifted here, the given signal is a simple sine wave. There is no attenuation or noise either, since the amplitudes remain uniform. Therefore, no transmission impairment is present.

Q2:
Bandwidth = f_{max} - f_{min}
f_{max} = Bandwidth + f_{min} = 20 + 25 = 45 kHz

Check figure 3.13 in the book, it shows how frequency domain plots are drawn for composite signals (both periodic and non-periodic), for **Q3** as well.

Q4:
This math is similar to Example 3.20 from the book.

Total number of pixels in a frame = 1920 x 1080

Each refresh produces a new frame

60 refreshes = 60 new frames = 60 x 1920 x 1080 pixels

1 frame = 24 bits

Total number of bits = 60 x 1920 x 1080 x 24 = 2985984000 bits

After compression, number of bits = 2985984000 x 5% = 149299200 bits

Q5:
Deterioration in 1 km = 3 W
Deterioration in 6 km = 3 x 6 W = 18 W

Signal strength at source, p_1 = 20 W
After attenuation, strength at 6 km from source, p_2 = 20 - 18 W = 2 W

Attenuation at 6 km from source, A_1 = $10\log_{10}$(p2/p1) = $10\log_{10}$(2/20) = -10 dB
Signal strength after 15x amplification, p_3 = 2 x 15 W = 30 W
Attenuation after amplifier, A_2 = $10\log_{10}(p_3/p_2)$ = $10\log_{10}$(30/2) = 11.76 dB

Deterioration at the endpoint (10 - 6 = 4 km from the amplifier) = 3 x 4 W = 12 W
Signal strength at the endpoint, p_4 = 30 - 12 W = 18 W

Attenuation at the endpoint, $A_3 = 10\log_{10}(p_4/p_3) = 10 \log_{10}(18/30) = -2.22$ dB

Total attenuation = $A_1 + A_2 + A_3 = (-10 + 11.76 - 2.22) = -0.46$ dB
(Note that this equals the attenuation calculated just by using the strength of the initial (20) and final (18) signals = $10\log_{10}(18/20)$ dB)

Q6.
a) 1 character requires 8 bits
1.35×10^8 characters require 10.8×10^8 bits

Number of characters sent in 5 minutes (5 x 60 = 300 seconds) = 10.8×10^8
Number of characters sent in 1 second = 3.6×10^6

Since this is 75% (¾) of the Shannon capacity,
Actual Shannon capacity, C = $(4/3) \times 3.6 \times 10^6 = 4.8 \times 10^6$ bps = 4.8 Mbps

b) $C = B \times \log_2(1+SNR)$
$SNR = 2^{C/B} - 1 = 2.81 \times 10^{14}$

c) $C = 2 \times B \times \log_2 L$ [here, C will be the actual bit rate i.e. 3.6×10^6]
$L = 2^{C/(2 \times B)} = 2^{18} = 262144$

Q7.

Frame size = 5 MB = $5 \times 10^6 \times 8$ bits (note that using 1024 x 1024 instead of 1000 x 1000 would have been more accurate for converting from MB to Bytes)

Throughput = 5 Mbps = 5×10^6 bps

Transmission delay = Frame Size/Throughput = 8 seconds

Propagation delay = Distance / Propagation Speed = $(2 \times 10^6)/(2 \times 10^8) = 0.01$ seconds

Queuing delay per router = 2 μs = 2×10^{-6} seconds

Total queuing delay for 10 routers = 2×10^{-5} seconds

Processing delay per router = 1 μs = 10^{-6} seconds

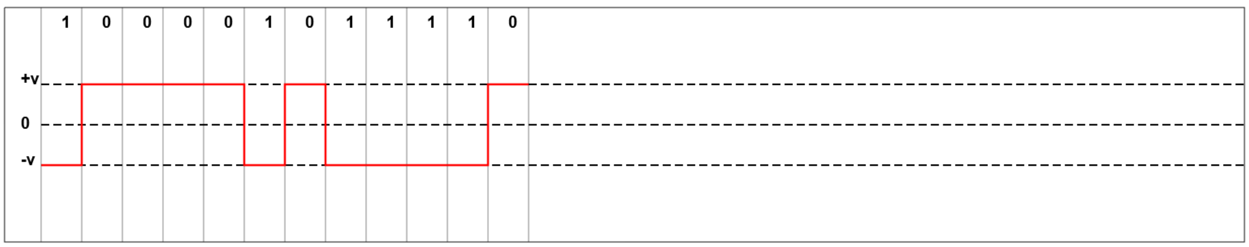
Total processing delay for 10 routers = 10^{-5} seconds

Transmission delay is dominant. Queuing delay and processing delay are negligible.

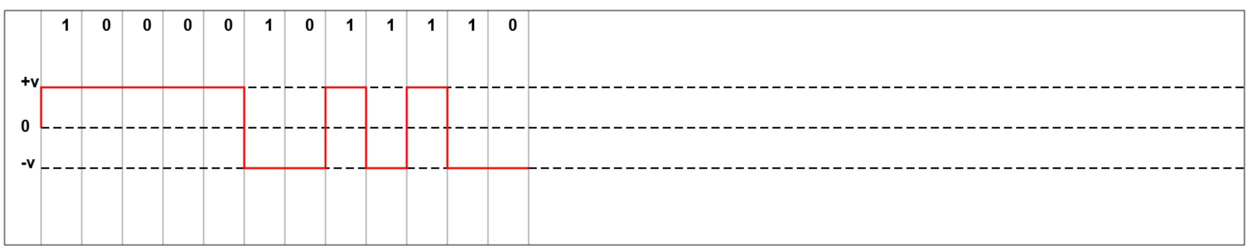
Total delay = transmission delay + propagation delay + queuing delay + processing delay
= 8.01003 seconds

Q8.

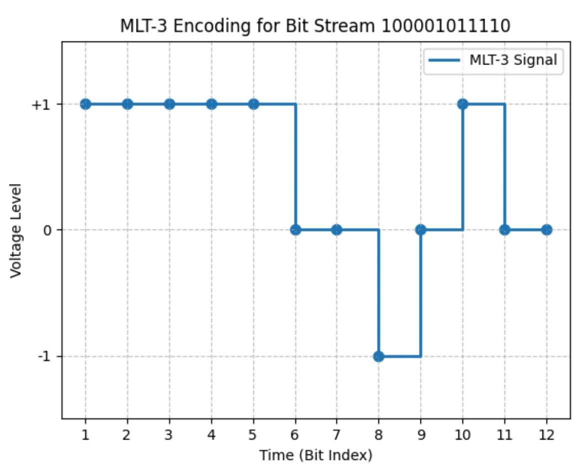
NRZ-L:



NRZ-I:



MLT-3:

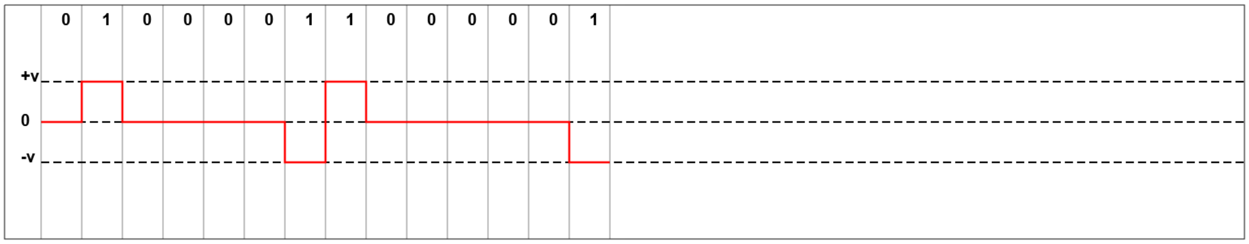


Q9.

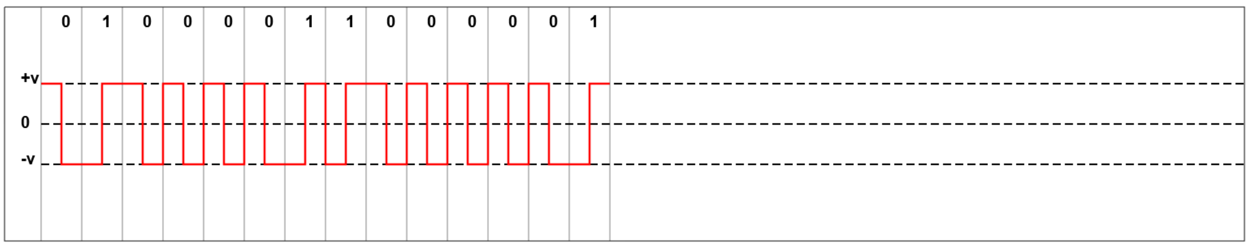
a) The given bit stream is 01000V11B00V01, which should be decoded to 01000011000001.

b) We can use bipolar AMI or the biphase schemes (Manchester or differential Manchester).

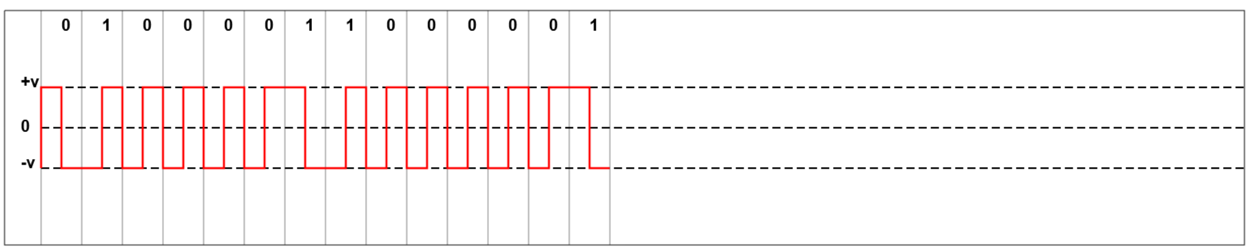
Bipolar AMI:



Manchester:



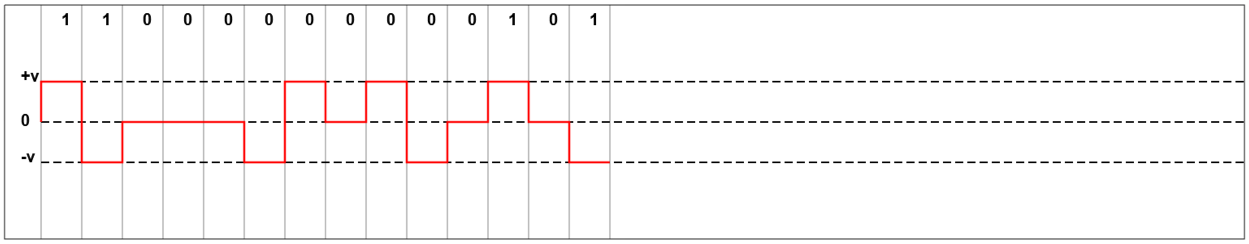
Differential Manchester:



In terms of bandwidth, bipolar AMI is better. In terms of self-synchronization, biphase schemes are better.

Q10.

a) The schemes that use 3 levels include AMI and MLT-3. MLT-3 has the staircase problem for long sequences of 1's (check figure 4.13). Therefore, AMI has been used here.
The decoded signal is 11000000000101.
b) This signal suffers from a lack of self-synchronization in case of a long sequence of 0's.
c) We can use B8ZS if the number of consecutive 0's is at least 8. Here, the number is 9, so we will use B8ZS.


1 1 0 0 0 0 0 0 0 0 0 1 0 1
+v
0
-v